

# VISHAL KOTHA

**Materials Chemistry | Nanofabrication | Electrocatalysis | Microscopy**  
 Email: [vishalkotha1@gmail.com](mailto:vishalkotha1@gmail.com) | Phone: +91-9488659155 | Location: Berhampur, Odisha, India  
 LinkedIn: [linkedin.com/in/vishal-kotha](https://www.linkedin.com/in/vishal-kotha) | Personal Website: [vishalkotha.com](http://vishalkotha.com)  
 Google Scholar: [scholar.google.com/citations](https://scholar.google.com/citations)

## PROFESSIONAL SUMMARY

Materials chemist with a Ph.D. in Chemistry from IIT Bombay and 7+ years of research and teaching-adjacent experience in nanomaterials, thin films, and electrocatalysis for clean energy applications (green hydrogen, batteries, solar cells). Currently leading an alkaline water electrolyzer project for green hydrogen production at BPCL-CRDC, with hands-on expertise in commercial OER/HER and AOR/HER electrocatalyst systems. Skilled in advanced characterization (TEM, SEM, XPS, XRD, AFM, electrochemical methods) and nanofabrication (electron-beam and photolithography, ALD, CVD). Published 11 peer-reviewed articles in journals including *J. Mater. Chem. A* and *ACS Appl. Energy Mater.* Experienced in mentoring graduate/undergraduate researchers and teaching assistantship in electrochemistry at the postgraduate level.

## PROFESSIONAL EXPERIENCE

2026 – Present



**Scientist-B (FTE)**, *Bharat Petroleum Corporation Limited*; Greater Noida, India

- Leading the alkaline electrolyzer project for green hydrogen production, with focus on commercial-scale OER/HER and AOR/HER electrocatalyst systems for industrial water electrolysis.
- Contributing to an electrochemical H<sub>2</sub>S valorization project, exploring sulfide-based anodic reactions as an alternative to conventional OER.
- Developing retardant catalyst materials for a ceramic hydrogen burner as part of process safety and combustion-control research.

2024 – 2025



**Teaching Volunteer**, *Isha Foundation*; Coimbatore, India

- Taught ICSE Chemistry to Grade 9 and 11 students (one semester) at Isha Home School and Science & Mathematics at the Isha Samskriti (one semester), under a structured residential teaching program.
- Conducted supplementary chess instruction for students at both institutions.

2023 – 2024



**Postdoctoral Fellow**, *Weizmann Institute of Science*; Rehovot, Israel

- Pioneered chirality induction in Ni-Au systems for hydrogen evolution reaction (HER), improving efficiency by 10% for green fuel applications, characterized via TEM and XPS.
- Collaborated with international teams, demonstrated superior HER activity, emphasizing sustainable catalysis potential (terminated early due to regional conflict).

2022 – 2023



**Postdoctoral Fellow**, *Indian Institute of Technology Bombay*; Mumbai, India

- Investigated magneto-electric effects on water electrolysis using La<sub>2</sub>FeMnO<sub>6</sub>, advancing multi-ferroic catalysts for energy conversion.
- Conducted mechanochemical shear-induced nucleation and exfoliation of nanomaterials, contributing to scalable synthesis techniques for industrial energy applications.

2016 – 2022



**PhD researcher**, *Indian Institute of Technology Bombay*; Mumbai, India

- Led hydrothermal synthesis of K-substituted LaMnO<sub>3</sub> perovskite for zinc-air batteries and organocatalysis, resulting in 3 high-impact publications.
- Operated and trained peers on characterization tools (FEG-TEM, SEM-EDS, PXRD, ICP-AES), logging 500+ hours and mentoring 4 students.
- Fabricated MOSFET devices using 1D nanorods, exploring interconnects for energy-efficient electronics.

## TEACHING EXPERIENCE

- Teaching Assistant, Electrochemistry (CH117L), Department of Chemistry, IIT Bombay (2016–2018) — 30/30 lab sessions conducted
- ICSE Chemistry (Grade 9 & 11), Isha Home School, Coimbatore (2024)
- Science & Mathematics (secondary level), Isha Samskriti, Coimbatore (2024–2025)
- Visiting Tutor in Chemistry, Dawn Breeze International School, Berhampur (2012–2013).

## EDUCATION

2016 – 2022


**Ph.D. in Chemistry, Indian Institute of Technology Bombay** | Mumbai, India

CPI: 8.44/10

**Thesis:** *Tailoring Transition Metal Perovskite Oxides via Low-Temperature Hydrothermal Routes as Potential Candidates for Catalytic Applications*
**Supervisor:** Prof. Leela Srinivas Panchakarla (late)

2014 – 2016


**M.Sc. in Chemical Sciences, Pondicherry University** | Puducherry, India

CGPA: 8.4/10

**Thesis:** *On the Origin of Chink in B<sub>4</sub>C Armor: Anatomy of a classical 2c-2e B-B bond*
**Supervisor:** Prof. Musiri M. Balakrishna Rajan

2011 – 2014


**B.Sc. in Chemistry, Khallikote Unitary University** | Odisha, India

Percentage in Hons: 85.4%

*Graduated with Gold Medal for outstanding performance.*

## PUBLICATIONS

\*Indicates corresponding author.

- Nishad, H. S.; **Kotha, V.**; Sarawade, P.; Chaskar, A. C.; Mane, S.; Lee, J.; Walke, P. S. Exchanging interlayer anions in NiFe-LDHs nanosphere enables superior battery-type storage for high-rate aqueous hybrid supercapacitors. *J. Mater. Chem. A* **2024**, 12 (16), 9494–9507. DOI: [10.1039/D4TA00299G](https://doi.org/10.1039/D4TA00299G).
- Gupta, S. P.; **Kotha, V.**; Walke, P. S.; Panchakarla, L. S. Hydrated WO<sub>3</sub>·0.33H<sub>2</sub>O nanorod for excellent anode materials enables high performance and long-cycle stability for aqueous Zn-ion batteries. *J. Power Sources* **2024**, 604, 234500. DOI: [10.1016/j.jpowsour.2024.234500](https://doi.org/10.1016/j.jpowsour.2024.234500).
- Nishad, H. S.; Gupta, S. P.; **Kotha, V.**; Magdum, V. V.; Gawade, V. V.; Patole, P. S.; Biradar, A. V.; Walke, P. S. Sodium-substituted tungsten oxide nanoflowers: An efficient electrode enhancing the pseudocapacitive storage in aqueous asymmetric supercapacitors. *ChemNanoMat* **2024**, 10 (2), e202300463. DOI: [10.1002/cnma.202300463](https://doi.org/10.1002/cnma.202300463).
- Kotha, V.\***; Karajagi, I.; Ghosh, P. C.; Panchakarla, L. S. Potassium-substituted LaMnO<sub>3</sub> as a highly active and exceptionally stable electrocatalyst toward bifunctional oxygen reduction and oxygen evolution reactions. *ACS Appl. Energy Mater.* **2022**, 5 (6), 7297–7307. DOI: [10.1021/acsaem.2c00823](https://doi.org/10.1021/acsaem.2c00823).
- Kotha, V.\***; Reddy, P. L.; Maddu, D.; Panchakarla, L. S. Importance of K substitution in LaMnO<sub>3</sub> nanostructures for the selective reduction of  $\alpha,\beta$ -unsaturated carbonyl compounds and nitroarenes. *ACS Appl. Nano Mater.* **2022**, 5 (12), 17482–17486. DOI: [10.1021/acsanm.2c03867](https://doi.org/10.1021/acsanm.2c03867).
- Kotha, V.**; Kumar, K.; Dayman, P.; Panchakarla, L. S. Doping with chemically hard elements to improve photocatalytic properties of ZnO nanostructures. *J. Clust. Sci.* **2022**, 33 (5), 1943–1950. DOI: [10.1007/s10876-021-02115-3](https://doi.org/10.1007/s10876-021-02115-3).
- Bhavani, B.; Chanda, N.; **Kotha, V.**; Reddy, G.; Basak, P.; Pal, U.; Giribabu, L.; Prasanthkumar, S. 1D alignment of Co(II) metalated porphyrin-naphthalimide based self-assembled nanowires for photocatalytic hydrogen evolution. *Nanoscale* **2022**, 14 (1), 140–146. DOI: [10.1039/D1NR06961F](https://doi.org/10.1039/D1NR06961F).

8. Nishad, H. S.; Gupta, S. P.; **Kotha, V.**; Patil, B. M.; Chakane, S. D.; Bute, M. G.; Gosavi, S. W.; Late, D. J.; Walke, P. S. Enhanced van-der-Waals separation in hydrated tungsten oxide nanoplates enables superior pseudocapacitive charge storage. *J. Alloys Compd.* **2022**, 914, 165227. DOI: [10.1016/j.jallcom.2022.165227](https://doi.org/10.1016/j.jallcom.2022.165227).
9. Bera, A.; Pathak, S. S.; **Kotha, V.**; Prasad, B. L. V. Lamellar bimetallic thiolates: Synthesis, characterization, and their utilization for the preparation of bimetallic chalcogenide nanocrystals through mechanochemical grinding. *Adv. Mater. Interfaces* **2021**, 8 (23), 2100898. DOI: [10.1002/admi.202100898](https://doi.org/10.1002/admi.202100898).
10. Sugathan, V.; Ghosh, B.; Harikesh, P. C.; **Kotha, V.**; Vashishtha, P.; Salim, T.; Yella, A.; Mathews, N. Synthesis of bismuth sulphoiodide thin films from single precursor solution. *Sol. Energy* **2021**, 230, 714–720. DOI: [10.1016/j.solener.2021.10.041](https://doi.org/10.1016/j.solener.2021.10.041).
11. Mrinalini, M.; Suman Krishna, J. V.; Krishna, N. V.; **Kotha, V.**; Panchakarla, L. S.; Prasanthkumar, S.; Giribabu, L. Photobleaching of triphenylamine-phthalocyanine entails mixed valence-state triggered self-assembled nanospheres. *J. Phys. Chem. C* **2018**, 122 (34), 19946–19952. DOI: [10.1021/acs.jpcc.8b05068](https://doi.org/10.1021/acs.jpcc.8b05068).

## KEY PROJECTS

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- **Chiral Ni-Au Electrocatalysts (Weizmann, 2023–2024):** Developed chiral metal systems for enhanced HER, supporting hydrogen production for green fuels, characterized via TEM and XPS.
- **K-Substituted LaMnO<sub>3</sub> Catalysts (IIT Bombay, 2017–2022):** Synthesized cost-effective perovskites for zinc-air batteries and organic reductions, published in ACS Appl. Energy Matter.
- **MOSFET Fabrication (IIT Bombay, 2019–2021):** Fabricated 1D nanorod interconnects using ALD and lithography, enhancing energy-efficient device performance.
- **Dynamic Hydrothermal Synthesis (IIT Bombay, 2016–2020):** Tuned La-based perovskite nucleation for energy applications, using seed-assisted methods, laying groundwork for scalable catalysis.

## TECHNICAL SKILLS

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- **Synthesis:** Hydrothermal, mechanochemical, solid-state, solution-based, electroplating, CVD.
- **Characterization:** TEM (HR, SAED, EDAX), SEM (EDS), XRD (Rietveld refinement), XPS, AFM, ICP-AES, CV, LSV, Impedance, Ellipsometry.
- **Fabrication:** Electron beam lithography, Photolithography, Sputtering, ALD
- **Software:** Python (data analysis), FullProf\_Suite, PANalytical HighScore+, VASP, Gaussian, OriginLab, ChemDraw, ImageJ.
- **Project Management:** Led focused projects, trained teams, managed lab operations.

## MENTORSHIP

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- Inspire Summer Fellow: Anshul Gupta (Jan–May **2019**).
- M.Sc. Thesis: Pawan Dayman (**2018–2019**); Bhaskar (Jan–Mar **2020**).
- B.Sc. Thesis: Divya Maddu (**2019–2020**).

## INVITED LECTURES

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- "Future of Chemistry: Exploration of the Possibilities" — Resource Person, Department of Chemistry, Khallikote Unitary University, Berhampur (31/03/**2023**)
- "Insights into the Research Journey from Pursuing a PhD to Expectations post-PhD" — Resource Person, Department of Chemistry, Khallikote Unitary University, Berhampur (23/02/**2024**)

## **POSITIONS OF RESPONSIBILITY**

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- Mess Secretary, Hostel 12/13/14, IIT Bombay (2018)
- Institute Hygiene Committee Member, Hostel Affairs Council, IIT Bombay (2018)
- FEG-TEM Assistant, IIT Bombay (2017–2022): Managed sample prep, imaging, and EDS for 500+ hours
- PXRD Assistant, IIT Bombay (2018–2022): Conducted Rietveld refinement for 20+ samples
- FESEM-EDS Assistant, IIT Bombay (2020–2023): Optimized surface and elemental analysis
- Trained 3 PhD students in TEM/SEM operation, enhancing lab efficiency
- Mentor, IIT Bombay (2018–2020): Guided 1 PhD, 2 M.Sc., 1 B.Sc. student

## **CONFERENCES & SYMPOSIA**

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- Poster, "Low-Temperature Synthesis of K-substituted  $\text{LaMnO}_3$ ," ICCFM-2018, S.N. Bose National Centre, Kolkata
- Poster, "The Chinks in  $\text{B}_4\text{C}$  Armor," 15<sup>th</sup> Indian Theoretical Chemistry Symposium, University of Hyderabad (2016)
- Participant, Graphene Symposium, IIT Bombay (2016)
- Participant, Bengaluru-India Nano Roadshow, IIT Bombay (2018)

## **AWARDS & FELLOWSHIPS**

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- GATE (Chemistry), Ministry of Education, Govt. of India — 2016 (97.06 percentile, AIR 440)
- JAM (Joint Admission Test for M.Sc.), Ministry of Education, Govt. of India — 2014 (AIR 1080)
- Gold Medalist, B.Sc. (Hons.) Chemistry, for academic excellence

## **REFERENCES**

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- **Bhagavatula L.V. Prasad**, Director, CeNS, Bengaluru, IN | [pl.bhagavatula@ncl.res.in](mailto:pl.bhagavatula@ncl.res.in)
- **Anindya Datta**, Professor, IIT Bombay, IN | [anindya@chem.iitb.ac.in](mailto:anindya@chem.iitb.ac.in)
- **Seelam Prasanth Kumar**, Scientist, CSIR-IICT, IN | [prasanth@iiict.res.in](mailto:prasanth@iiict.res.in)